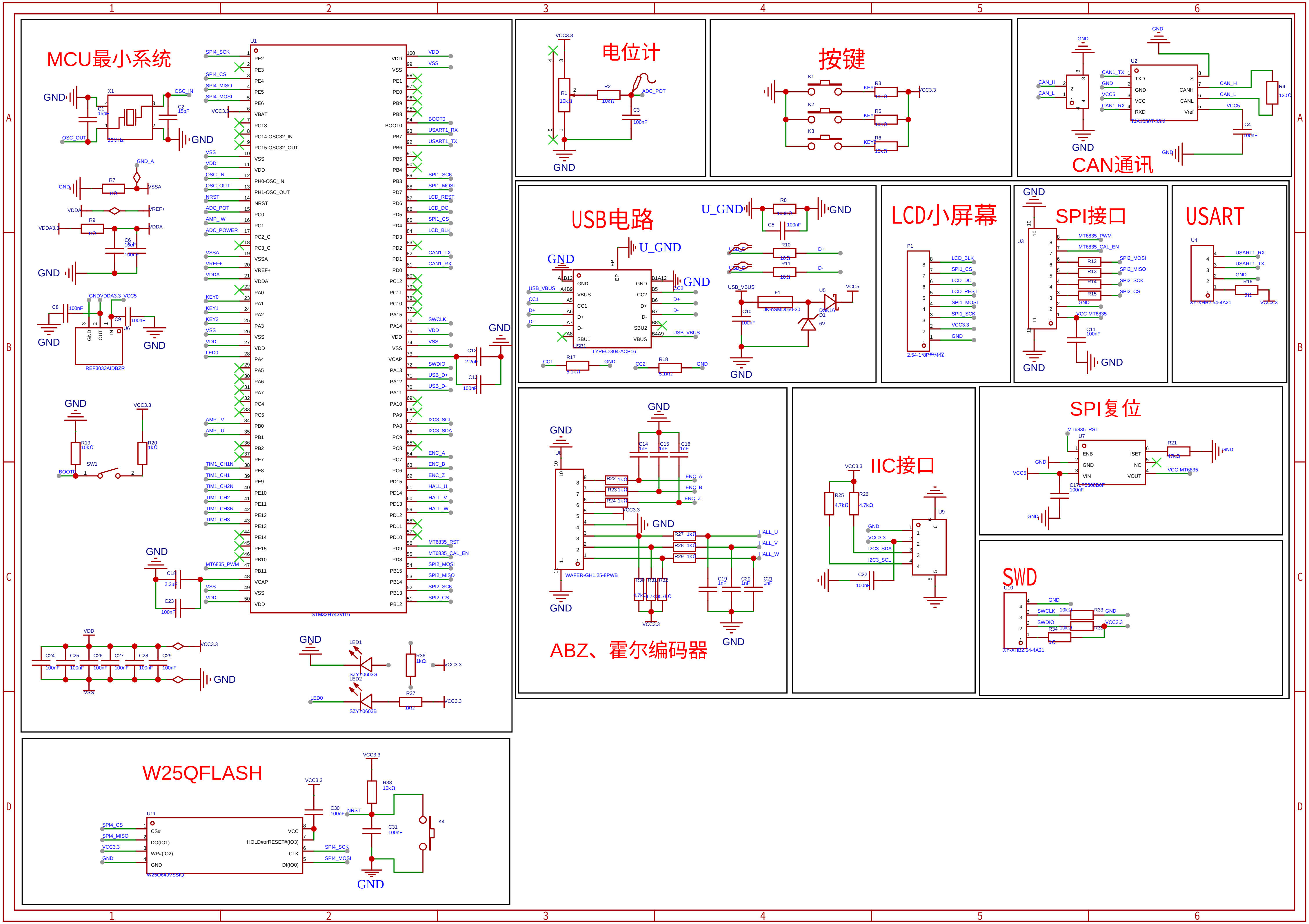
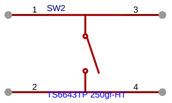
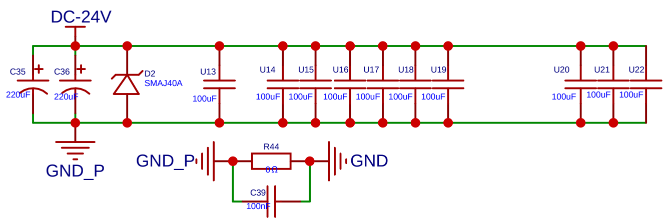


MCU最小系统
电位计
按键
CAN通讯
USB电路
LCD小屏幕
SPI接口
USART
ABZ、霍尔编码器
IIC接口
SPI复位
SWD
W25QFLASH
U1
STM32H743VIT6
X1
25MHz
C1 15pF
C2 15pF
OSC_IN
OSC_OUT
GND
VCC3.3
VDD
VSS
VSSA
VDDA
VREF+
VDDA3.3
R7 0Ω
R9 0Ω
GND_A
C6 1uF
100nF
C8 100nF
C9 100nF
U6
REF3033AIDBZR
VCC5
R19 10kΩ
R20 1kΩ
SW1
BOOT0
C18 2.2uF
C23 100nF
C12 2.2uF
C13 100nF
C24 C25 C26 C27 C28 C29
100nF
LED1
LED2
SZYY0603G
SZYY0603B
R36 1kΩ
R37 1kΩ
LED0
R1 10kΩ
R2 10kΩ
C3 100nF
ADC_POT
K1 K2 K3
KEY0 KEY1 KEY2
R3 10kΩ
R5 10kΩ
R6 10kΩ
U2
TJA1050T-3/5M
TXD GND VCC RXD S CANH CANL Vref
CAN1_TX CAN1_RX
CAN_H CAN_L
R4 120Ω
C4 100nF
U_GND
R8 100kΩ
C5 100nF
R10 10Ω
R11 10Ω
USB_D+ USB_D-
D+ D-
USB1
TYPEC-304-ACP16
USB_VBUS
CC1 CC2
SBU1 SBU2
R17 5.1kΩ
R18 5.1kΩ
F1
JK-nSMD050-30
D1
DSK16
6V
U5
C10 100nF
P1
2.54-1*8P母针保
LCD_BLK SPI1_CS LCD_DC LCD_REST SPI1_MOSI SPI1_SCK
U3
MT6835_PWM
MT6835_CAL_EN
SPI2_MOSI SPI2_MISO SPI2_SCK SPI2_CS
R12 R13 R14 R15
VCC-MT6835
C11 100nF
U4
XY-XHB2.54-4A21
USART1_RX USART1_TX
R16 0Ω
U8
WAFER-GH1.25-8PWB
R22 1kΩ
R23 1kΩ
R24 1kΩ
ENC_A ENC_B ENC_Z
C14 1nF
C15 1nF
C16 1nF
R27 1kΩ
R28 1kΩ
R29 1kΩ
HALL_U HALL_V HALL_W
R30 R31 R32 4.7kΩ
C19 1nF
C20 1nF
C21 1nF
R25 4.7kΩ
R26 4.7kΩ
U9
I2C3_SDA I2C3_SCL
C22 100nF
U7
MT6835_RST
ENB GND VIN ISET NC VOUT
R21 47kΩ
C17 100nF
LP5900B6F
U10
XY-XHB2.54-4A21
SWCLK SWDIO
R33 10kΩ
R34 10kΩ
R35 0Ω
U11
W25Q64JVSSIQ
CS# DO(IO1) WP#(IO2) GND VCC HOLD#orRESET#(IO3) CLK DI(IO0)
SPI4_CS SPI4_MISO SPI4_SCK SPI4_MOSI
C30 100nF
R38 10kΩ
C31 100nF
NRST
K4
1 2 3 4 5 6
A B C D

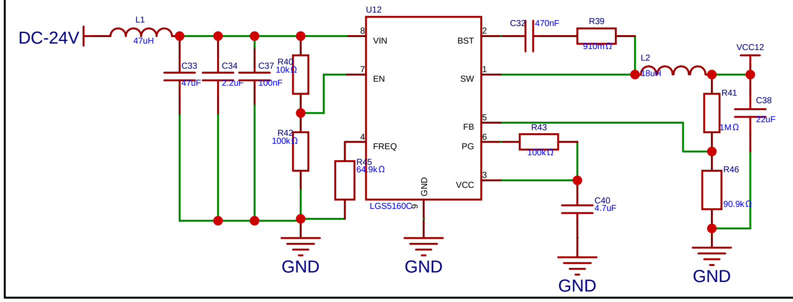


SW2
TS6643TP 250gf-H1
1 2 3 4

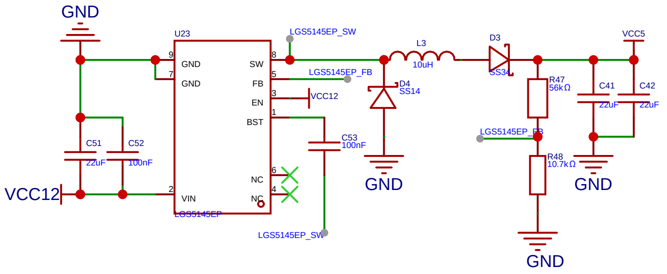
母线电源滤波


DC-24V
C35
2200uF
C36
220uF
D2
SMAJ40A
U13
100uF
U14
100uF
U15
100uF
U16
100uF
U17
100uF
U18
100uF
U19
100uF
U20
100uF
U21
100uF
U22
100uF
GND_P
GND_P
R44
0Ω
GND
C39
100nF

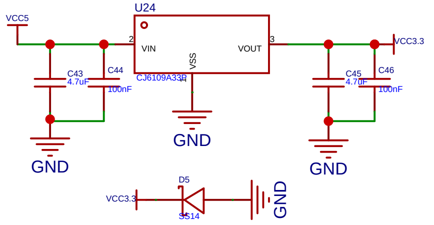
DCDC 24V to 12V


U12
DC-24V
L1
47uH
C33
47uF
C34
2.2uF
C37
100nF
R40
10kΩ
R42
100kΩ
R45
64.9kΩ
8
VIN
7
EN
4
FREQ
GND
LGS5160C
9
2
BST
1
SW
5
FB
6
PG
3
VCC
C32
470nF
R39
910mΩ
L2
18uH
VCC12
R41
1MΩ
C38
22uF
R43
100kΩ
R46
90.9kΩ
C40
4.7uF
GND
GND
GND
GND

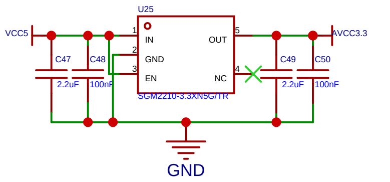
DCDC 12V to 5V


GND
U23
9
GND
7
GND
2
VIN
LGS5145EP
8
SW
5
FB
3
EN
1
BST
6
NC
4
NC
LGS5145EP_SW
LGS5145EP_FB
VCC12
C53
100nF
LGS5145EP_SW
C51
22uF
C52
100nF
VCC12
L3
10uH
D3
SS34
D4
SS14
GND
R47
56kΩ
LGS5145EP_FB
R48
10.7kΩ
GND
C41
22uF
C42
22uF
VCC5
GND

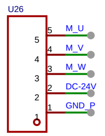
LDO 5V to 3.3V

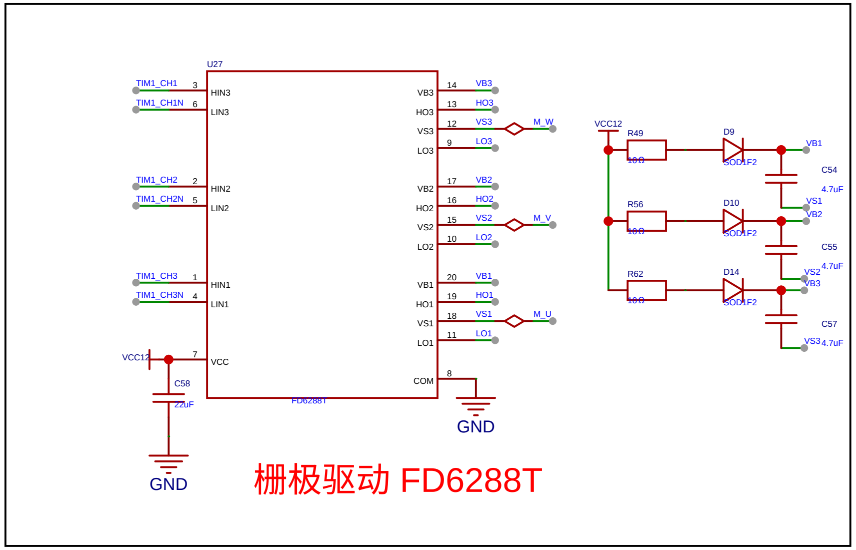

U24
VCC5
2
VIN
VSS
1
VOUT
3
CJ6109A33P
C43
4.7uF
C44
100nF
GND
GND
C45
4.7uF
C46
100nF
VCC3.3
GND
D5
VCC3.3
SS14
GND

LDO 5V to 3.3V 用于运放


U25
VCC5
1
IN
2
GND
3
EN
5
OUT
4
NC
SGM2210-3.3XN5G/TR
C47
2.2uF
C48
100nF
C49
2.2uF
C50
100nF
AVCC3.3
GND

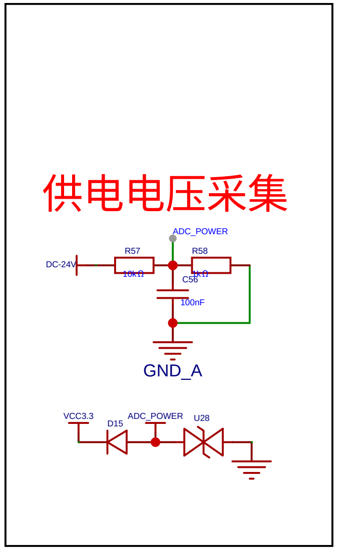
电源和三相接口


U26
5
4
3
2
1
M_U
M_V
M_W
DC-24V
GND_P



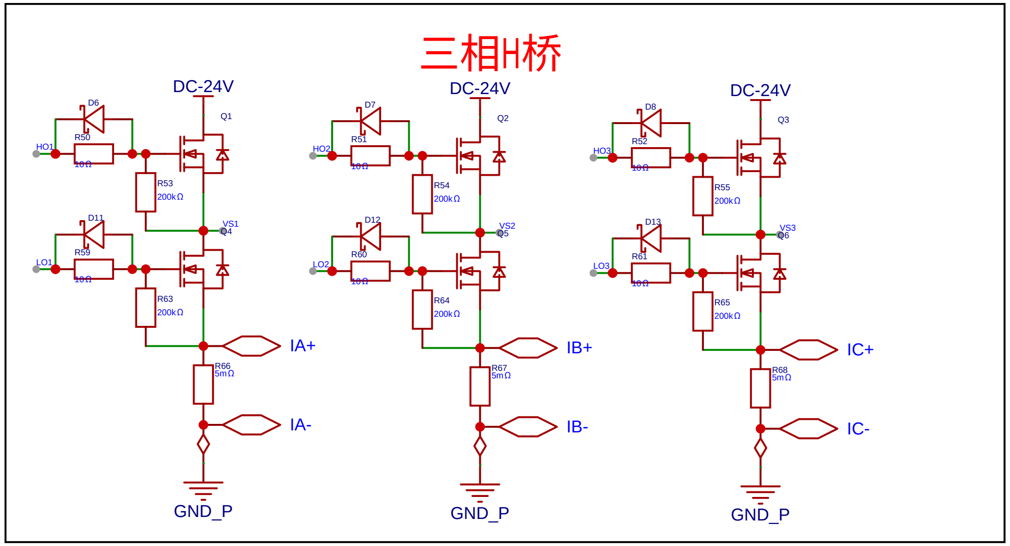
栅极驱动 FD6288T

U27
FD6288T
TIM1_CH1
TIM1_CH1N
TIM1_CH2
TIM1_CH2N
TIM1_CH3
TIM1_CH3N
HIN3
LIN3
HIN2
LIN2
HIN1
LIN1
VCC
VB3
HO3
VS3
LO3
VB2
HO2
VS2
LO2
VB1
HO1
VS1
LO1
COM
M_W
M_V
M_U
VCC12
C58
22uF
GND
R49
R56
R62
10Ω
D9
D10
D14
SOD1F2
C54
C55
C57
4.7uF
VB1
VS1
VB2
VS2
VB3
VS3



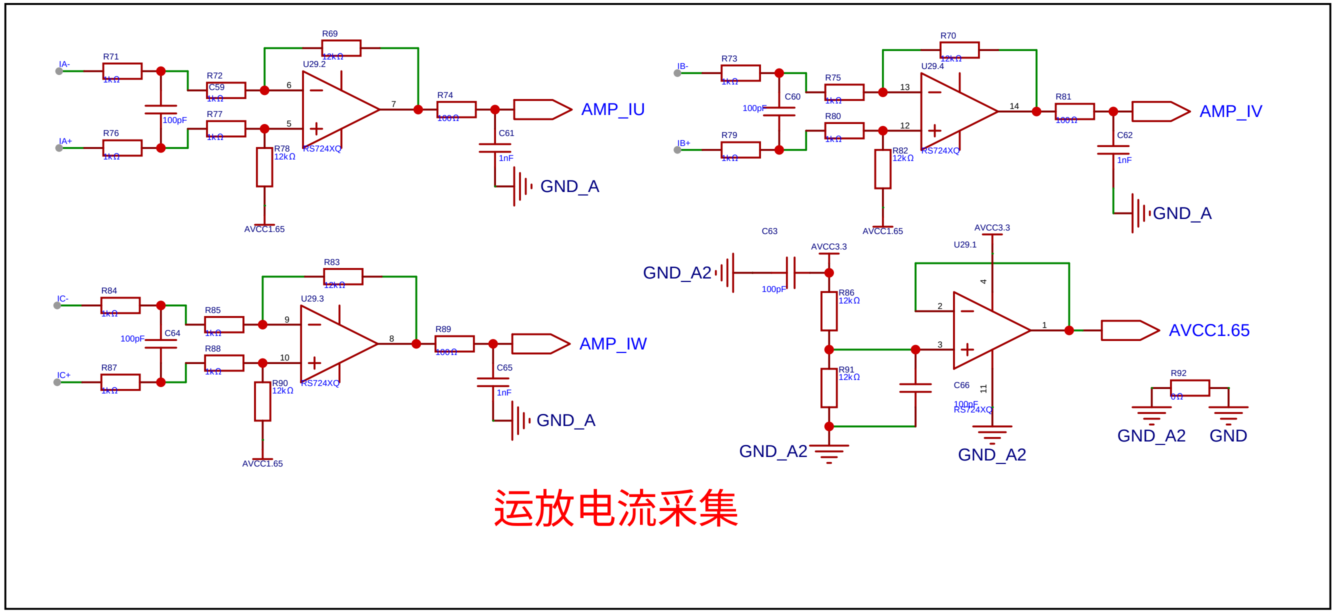
供电电压采集

DC-24V
R57
10kΩ
R58
1kΩ
ADC_POWER
C56
100nF
GND_A
VCC3.3
D15
ADC_POWER
U28



三相H桥

DC-24V
Q1
Q2
Q3
Q4
Q5
Q6
D6
D7
D8
D11
D12
D13
HO1
HO2
HO3
LO1
LO2
LO3
R50
R51
R52
R59
R60
R61
10Ω
R53
R54
R55
R63
R64
R65
200kΩ
VS1
VS2
VS3
IA+
IA-
IB+
IB-
IC+
IC-
R66
R67
R68
5mΩ
GND_P



运放电流采集

IA-
IA+
IB-
IB+
IC-
IC+
R71
R76
R72
R77
R73
R79
R75
R80
R84
R87
R85
R88
1kΩ
R69
R70
R83
12kΩ
R78
R82
R90
12kΩ
C59
C60
C64
100pF
U29.2
U29.4
U29.3
U29.1
RS724XQ
R74
R81
R89
100Ω
C61
C62
C65
1nF
AMP_IU
AMP_IV
AMP_IW
GND_A
AVCC1.65
C63
100pF
GND_A2
AVCC3.3
R86
12kΩ
R91
12kΩ
C66
100pF
AVCC1.65
R92
0Ω
GND_A2
GND